Mainul Islam

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Education

B.Sc. in Electrical and Electronic Engineering Jan 2020 – Jun 2024
Islamic University of Technology (IUT), Gazipur, Bangladesh CGPA: 3.87 / 4.00

Thesis -

- MonoSIM, an open-source SIL platform developed for testing autonomous control algorithms on Ackermann vehicles.
- The system was designed using open source tools, creating an environment with a monocular camera vision system to capture stimuli from it with minimal computational overhead through a sliding window based lane detection method.
- The system is validated with PID vs. MPC controllers, showing PID had lower tracking error while MPC gave smoother, constraint-aware control.

Professional Experience

Research and Development Engineer — Cybernetics Hi-Tech Solutions Pvt. Ltd., Dhaka Oct 2024 – Present

- Contributed to the development of robotic and embedded systems from initial requirements through to delivery, covering hardware, software, and operator interfaces, for applications like industrial-automation etc.
- Design and implemented autonomous navigation stack for ground vehicles, including positioning, route planning, motion control, and the coordination of multi-vehicle fleets.
- Developed real-time embedded system for distributed motor control and sensor systems across a range of industrial communication standards.
- Designed custom control electronics from schematic through fabrication, testing, and commissioning for demanding field environments.

Lecturer (Part-Time) — Ahsanullah University of Science and Technology, Dhaka Jul 2025 – Jun 2026

- Conducted C++ Programming and Digital Signal Processing courses.
- Delivered lectures, prepared laboratory manuals, and conducted student assessment.

Publications

Peer-Reviewed Journal Articles

- A. U. R. Adib, **M. Islam**, M. S. Abid, R. Ahshan, "A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI," *Solar Energy*, vol. 302, 2025. (Impact Factor: 6.6) DOI: 10.1016/j.solener.2025.114058
- S. Rahman, A. Banik, M. K. Hossain, **M. Islam**, A. Al-Hysam, I. Ahmed, "Topological survey of zeta power converters from classical designs to emerging architectures," *Energy Conversion and Management: X*, art. no. 102222, 2026. (Impact Factor: 8.8) DOI: 10.1016/j.ecmx.2026.102222

Peer-Reviewed Conference Papers

- S. Rahman, N. Hasin, **M. Islam**, M. Z. A. Rony, G. Sarowar, "MonoSIM: An open-source SIL framework for Ackermann vehicular systems with monocular vision," *IEEE 12th Intl. Conf. on Automation, Robotics and Applications (ICARA)*, 2026. DOI: 10.1109/ICARA69401.2026.11480327
- N. Hasin, M. A. R. Nishat, **M. Islam**, K. S. A. Hasan, A. Newaz, "A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR," *IEEE Intl. Conf. on AI and Data Analytics (ICAD)*, 2026. DOI: 10.1109/ICAD69378.2026.11608728
- H. Haque, **M. Islam**, M. A. I. Alvy, M. S. Islam, M. A. Razzak, "Long-term energy demand analysis using machine learning algorithms: A case study in Bangladesh," *IEEE ICECIE*, 2024. DOI: 10.1109/ICECIE63774.2024.10815643
- M. S. Tanvir, F. A. Tonmoy, M. M. Islam, **M. Islam**, M. Hasan, "Optimizing master–slave broadcast and unicast communication in multi-node STM32 networks using UART and RS485 protocols," *IEEE ICCIT*, 2024. DOI: 10.1109/ICCIT64611.2024.11022430

Manuscripts Under Review

- M. M. Islam, **M. Islam**, M. S. Tanvir, “Speed synchronization of a dual-BLDC tracked differential-drive vehicle using a GA-tuned multi-rate cross-coupled controller.”

Projects

Man-Transportable EOD Robot (Jontro Soinik 2.0)

- Developed the motion-control library governing eleven servo drives distributed over two independent CAN buses, implementing the CANopen (CiA-402) standard so that each joint can be commanded by position, speed, or torque.
- Contributed to a closed-form solution to the arm’s inverse kinematics. Verified against the robot’s geometric model and integrated with MoveIt 2 for collision checking.
- Wrote the STM32 firmware carrying the operator’s control signal over an Ethernet link via optical fiber.
- Developed an app interface, presenting low-latency video feeds, a live telemetry dashboard, and a three-dimensional on-screen model mirroring the robot’s actual posture.

Autonomous Guided Vehicle — CYBERFLEET

- Designed the complete ROS 2 system integration for a three-vehicle warehouse robot fleet, covering how each vehicle establishes its position, plans a route, follows it, and coordinates with the others.
- Implemented an extended Kalman filter combining continuous wheel-encoder motion with intermittent camera sightings of fixed floor markers.
- Built the marker-based localisation pipeline: camera calibration, pose estimation from each observed marker.
- Contributed to the route planning that treats time as a planning dimension, preventing both collisions and deadlocks.
- Implemented LiDAR-based obstacle detection.

Autonomous Target-Tracking Drone

- Contributed to the implementation of live onboard target detection & tracking, running a Pi 5 with an AI accelerator.
- Created the aerial training dataset and tested different tracking methods.

Toll-Booth Automation

- Developed the lane-controller firmware on an STM32 microcontroller, structured as a table-driven state machine.
- Integrated inductive loop coils, an infrared exit sensor, and licence-plate recognition to detect vehicles.
- Designed three custom circuit boards with optical isolation, surge and reverse-polarity protection.

Technical Skills

Programming: Python, C/C++, Embedded C, MATLAB.
Robotics and Control: ROS 2, sensor fusion, vision-based perception.
Embedded Systems: STM32 platforms, CAN, UART, Modbus.
Machine Learning: PyTorch, TensorFlow, scikit-learn, object detection, tracking, explainable AI.
Design Tools: EAGLE, KiCAD, easy eda.

Honours and Achievements

- **International Rover Challenge 2024:** 6th globally and 1st in Bangladesh; awarded ABEX Best Science Team.
- **European Rover Challenge 2023:** 16th globally (Project Altair).
- **MATLAB Cody:** Global rank 117, with 914 problems solved and 26 badges.

References

Prof. Dr. Golam Sarowar — B.Sc. Thesis Supervisor
Professor, Department of EEE, Islamic University of Technology (IUT)
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